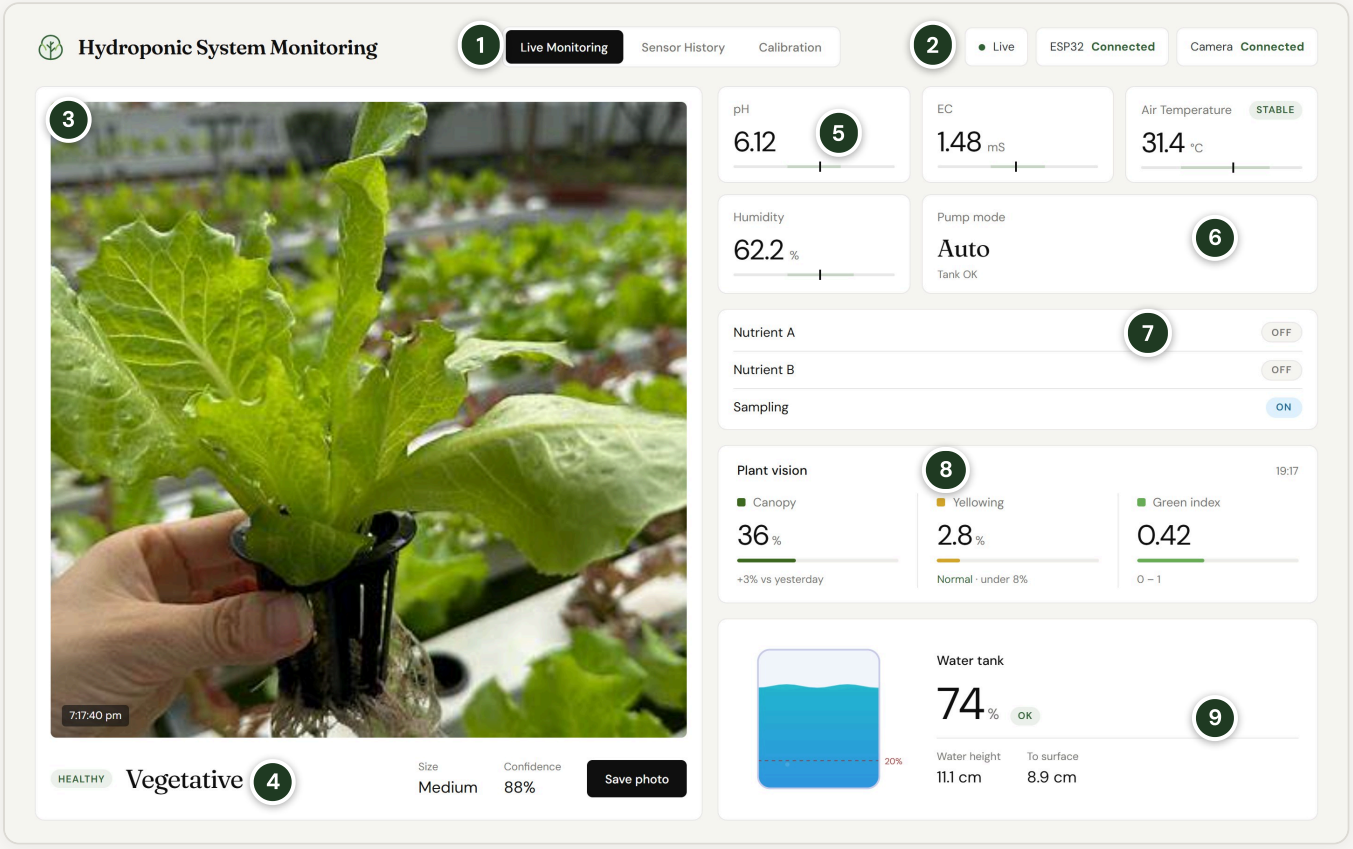
05

Dashboard Overview

A tour of the offline web dashboard: Live Monitoring, Sensor History and Calibration. It runs on the Raspberry Pi 5 and opens full screen on the HDMI monitor at boot.

TAB 1 · LIVE MONITORING

What is happening right now



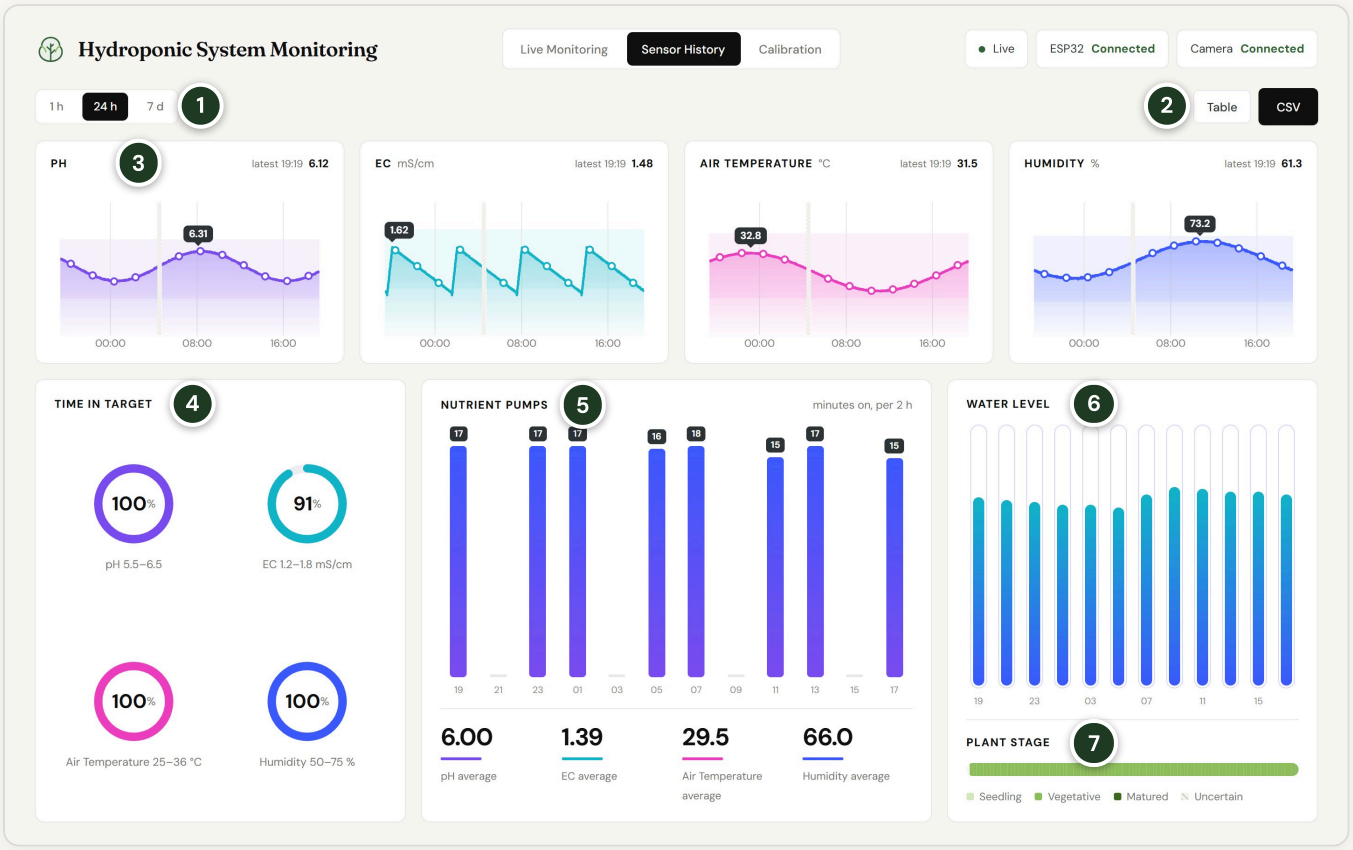
Screenshot taken 13 Nov 2026 by running the dashboard on a PC with simulated sensor data and a dataset photo; on the rack the same page shows live values.

1. **Tabs**: Live Monitoring, Sensor History, Calibration.
2. **Status chips**: MQTT link, ESP32 and camera connected or not.
3. **Live plant photo** from the MaixCAM, refreshed every 10 s.
4. **Plant result**: health, growth stage, size, AI confidence, **Save photo**.
5. **Sensor tiles**: pH, EC, air temperature (Cold / Stable / Hot) with the target band.
6. **Pump mode**: Auto or Manual, and whether the tank is OK.
7. **Pumps**: Nutrient A, Nutrient B, Sampling, each ON / OFF.
8. **Plant vision**: canopy %, yellowing %, green index.
9. **Water tank**: level %, the 20 % low line, water height, distance to surface.

TAB 2 · SENSOR HISTORY

How the system has been doing

History comes from the CSV session logs on the Pi through `server.py`. Pick a range, read the trends, export the data.



1. **Range**: last 1 h (every 2 s), 24 h (1-min averages) or 7 days (10-min averages).
2. **Table** shows the numbers; **CSV** downloads the log rows for the range.
3. **Charts**: pH, EC, air temperature, humidity. The shaded band is the target; hatched gaps mean no data. Hover one chart to see the same moment on all four.
4. **Time in target**: share of the range each value spent inside its band.
5. **Nutrient pumps**: minutes ON per time slot, plus the average of each sensor.
6. **Water level** over the range.
7. **Plant stage** strip from the camera log: seedling, vegetative, matured or uncertain.

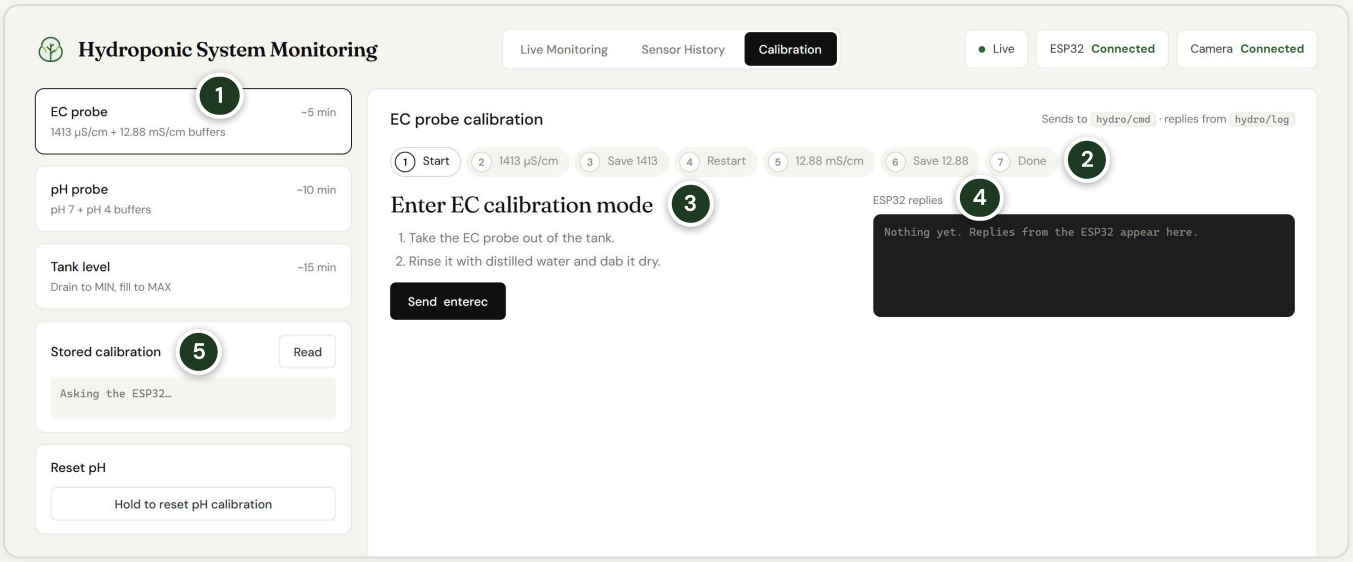
Target bands used for shading

Value	Target band	Chart scale	Colour
pH	5.5 – 6.5	5.0 – 7.0	violet
EC	1.2 – 1.8 mS/cm	0.8 – 2.0	cyan
Air temperature	25 – 36 °C	20 – 40	pink
Humidity	50 – 75 %	40 – 85	blue

TAB 3 · CALIBRATION

Guided sensor calibration

Step-by-step calibration for the EC probe, the pH probe and the tank level. Commands go to the ESP32 on `hydro/cmd`; its replies appear on the page.



1. **Choose a sensor**: EC (~5 min), pH (~10 min) or tank level (~15 min).
2. **Step bar**: e.g. EC: Start → 1413 µS/cm → Save → Restart → 12.88 mS/cm → Save → Done.
3. **Instructions + send button**. Capture buttons stay disabled until the reading is steady.
4. **ESP32 replies**, e.g. `# EC K values written to flash`.
5. **Stored calibration** (`status`) and a hold-to-reset for pH.

Opening the dashboard

Where	How
HDMI monitor on the Pi	Opens by itself at boot (Firefox kiosk, screen never blanks)
Browser on the Pi	`http://localhost:8000`
Phone / laptop	`http://<pi-ip>:8000` on the same network

If something looks wrong

On screen	Meaning / fix
"Live link down (MQTT :9001)"	Browser cannot reach Mosquitto's websocket. Tiles and calibration need it; the photo still updates over HTTP.
Greyed-out value	A held value: the last good reading, kept up to 20 s while the sensor misses a read.
Camera not connected	Check the USB-C cable and that the MaixCAM shows the Hydro Cam preview (`ping 10.122.173.1`).

> The Calibration tab can send commands to the ESP32 and MQTT has no password yet, so anyone on the same network could do the same. Add a Mosquitto password before using the dashboard on a shared network.